

MLFF PERF-P3 CPU Benchmark Evidence

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Scope

PERF-P3 is an execution-equivalent CPU hardening gate. This report records matched CPU evidence for the direct local-structure kernel, exact FOUNDATION-AUDIT1 temporary-memory handling, and rejected candidate optimizations. Accelerator evidence is intentionally deferred to FINAL-GPU1.

Host

The bounded cloud host exposed an AMD EPYC 9V74 processor, nine logical CPUs, an 8-core cgroup quota, and a 4 GiB memory limit. The structural microbenchmark fixes native BLAS/OpenMP work to one thread per fresh process so it measures the per-worker kernel rather than scheduler-dependent nested parallelism.

Direct structural kernel

The fixture uses a deterministic triclinic, fixed-topology, LTA-like system with 168 atoms and 300 frames. Three fresh-process samples were collected for each path.

Metric	0.20.184a0	PERF-P3	Change
Median wall time	3.458792 s	3.202054 s	7.42% lower
Wall-time range	3.331898-3.676044 s	3.193411-3.390388 s	-
Median peak RSS	270.35 MiB	270.33 MiB	effectively unchanged

Every sample produced the exact same numerical digest:

5786409f8f622b3e2d1183bcca9ef9859e3cfd7717970339f421d4f630d6ac4c

The qualified change therefore removes wrapper/allocation overhead without changing feature bytes.

Foundation-audit temporary memory

The audit fixture contains 600 frames with 1,500 atoms each, for 900,000 atoms total. It exercises the actual exact force-tail reduction while replacing prediction I/O with deterministic in-memory inputs.

Path	Wall time	Peak RSS	Scientific digest
Pre-P3 list + concatenate	0.157148 s	346.18 MiB	identical
P3 exact preallocation	0.192061 s	318.41 MiB	identical
P3 forced mmap	0.206037 s	318.46 MiB	identical

Preallocation reduces measured peak RSS by **8.02%**. It is not promoted as a speed optimization: the measured in-memory wall time is 22.22% higher on this deliberately reduction-heavy microfixture. The mmap path is a

bounded-allocation fallback; quantile evaluation still touches mapped pages, so mmap is not claimed to reduce resident memory further in this fixture.

All three paths produced:

a618301c7f8dad6f5cd8cfec4d39edea9303b5c2f1c52bf46b81507677808f9f

Rejected optimizations

Two candidates were deliberately rejected. Retaining dense reusable pair/radial scratch increased RSS and reduced throughput. Chunking the radial-channel reduction changed FP64 values at approximately 10^{-16} to 8.9×10^{-16} , so it was rejected even though the differences are numerically tiny: PERF-P3 is Class E and requires exact scientific identity.

Interpretation

The bounded authority supports a **7.42% median per-worker structural-kernel wall-time reduction** and an **8.02% audit peak-RSS reduction**. It does not support a GPU claim, a full-corpus end-to-end speedup claim, or a claim that mmap lowers RSS under every access pattern.